INDEX

Sr. No.	Name of Experiment	Date of Experiment	Date of Submission	Remarks
1(a)	Simple Calculator Application Using Servlet			
1(b)	Login Page Using Servlet			
1(c)	Registration Servlet Using JDBC			
2(a)	Request Dispatcher for Password Validation			
2(b)	Cookies to Store Number of Visits			
2(c)	Session Creation and Destruction			
3(a)	Servlet Application for File Upload and Download			
3(b)	Servlet Question Answer Application Using Database			
3(c)	Servlet Non-Blocking Read Operation			
4(a)	JSP Application Using Intrinsic Objects			
4(b)	JSP Page-to-Page Data Passing with Validation			
4(c)	JSP Registration and Login Using JDBC			
5(a)	Servlet Application for File Upload and Download			
5(b)	Servlet Question Answer Application Using Database			
5(c)	Servlet Non-Blocking Read Operation			
6(a)	JSP Employee Table Update Using Database			
6(b)	JSP Expression Language (EL)			
6(c)	JSP Standard Tag Library (JSTL)			
7(a)	Currency Converter Using EJB			
7(b)	Room Reservation System Using EJB			
7(c)	Shopping Cart Using Stateful Session Bean			
8(a)	Servlet Hit Count Using Singleton Session Bean			
8(b)	Visitor Statistics Using Message Driven Bean			
8(c)	Marks Entry Application Using EJB			
9(a)	Inventory Application Using JPA			
9(b)	Guestbook Application Using JPA			
9(c)	Book Details Using JPA			
10(a)	JPA Application Using ORM Associations			
10(b)	Hibernate Website Visitor Feedback in MySQL			
10(c)	Hibernate Employee Details in MySQL			
11(a)	Hibernate One-to-One Mapping Using Annotation			
11(b)	Hibernate Course Details with ORM Mapping			
11(c)	Five Page Web Application Using Java EE Technologies			
12(a)	Web Application Using Spring Boot 3			
12(b)	Application Using Spring Framework and Lightweight Containers			
12(c)	Dependency Injection with Spring			

Certificate

Class _____ **Year** _____

This is to certify that the work entered in this Journal is the work of

Shri / Kumari _____

of _____ Division _____ Roll No. _____

Uni. Exam No. _____ has satisfactorily completed the

required number of practical and worked for the 1st term / 2nd term/

both the terms of the year 20 _____ 20 _____ in the college laboratory

as laid down by the university.

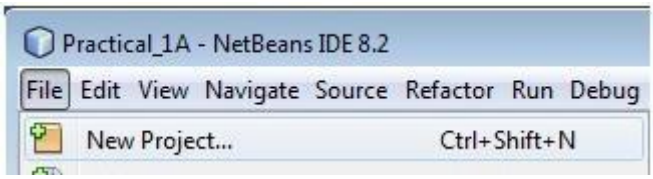
_____	_____	_____
Head of the Department	External Examiner	Internal Examiner Subject Teacher

Date : _____ / _____ / 20 Department of

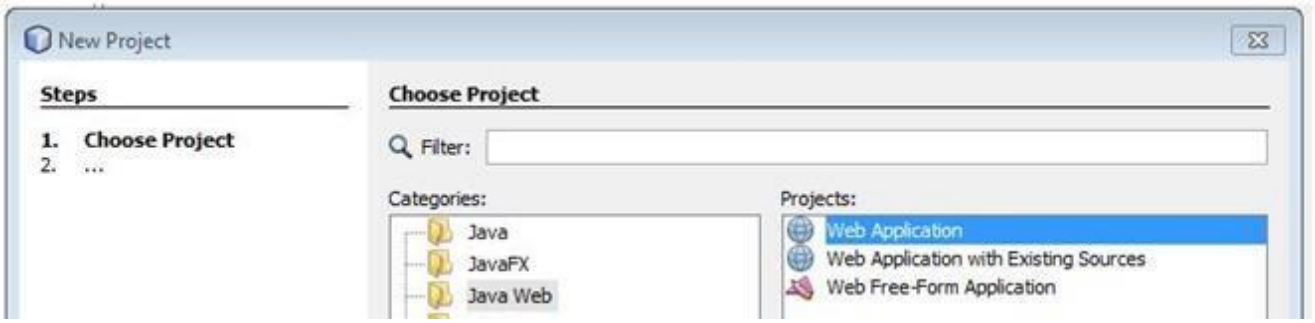
Requirements:

1. JDK 8u181
2. Netbeans 8.1 or higher
3. MySql 5.5 or higner

1a. Create a simple calculator application using servlet.



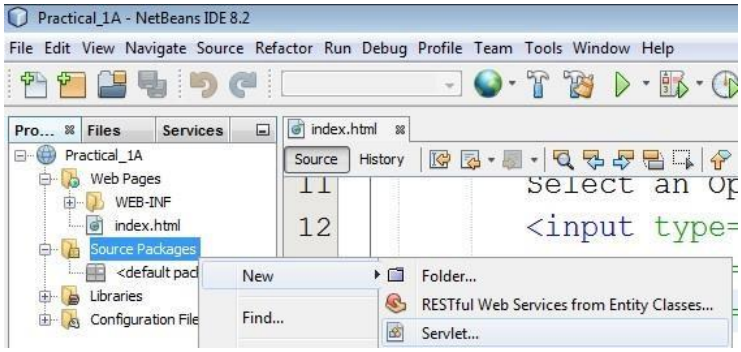
Create a new project



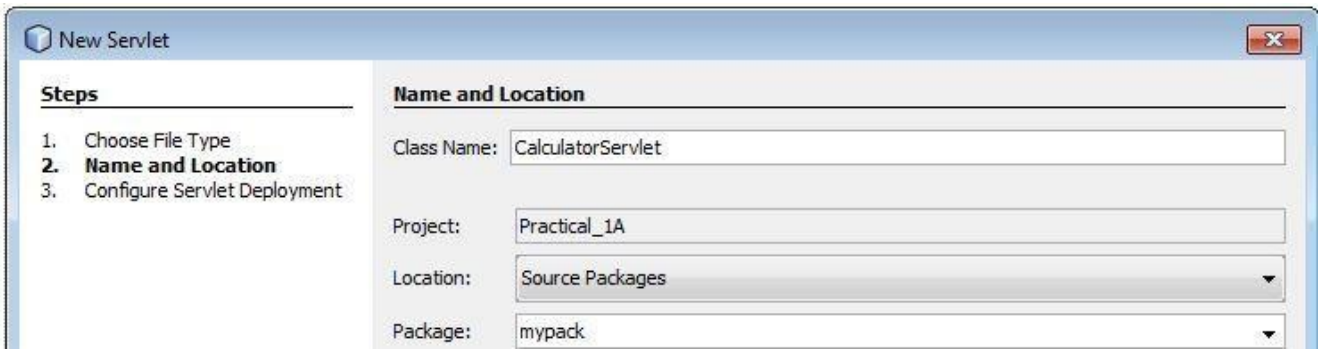
Select java web → web Application

index.html

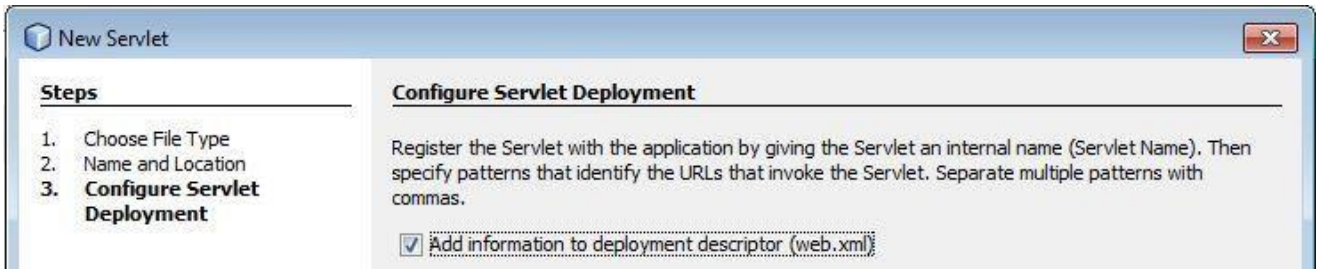
```
<html><head><title>Calculator App</title></head><body>
<form action="CalculatorServlet" >
    Enter First Number        <input type="text" name="txtN1" ><br>
    Enter Second Number  <input type="text" name="txtN2" ><br>
    Select an Operation
<input type="radio" name="opr" value="+">ADDTION
<input type="radio" name="opr" value="-">SUBSTRACTION
<input type="radio" name="opr" value="*">MULTIPLY
<input type="radio" name="opr" value="/">DIVIDE <br>
<input type="reset">
<input type="submit" value="Calculate" >
</form></body></html>
```



Add a new Servlet to source package



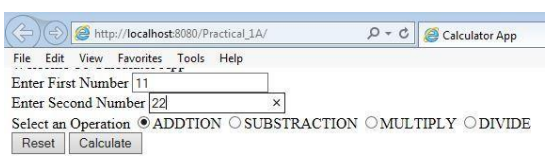
Name the servlet and package

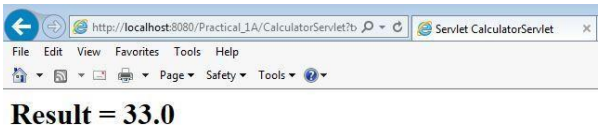


Check "Add information to deployment descriptor" to add servlet into web.xml
But preferably use annotation.

CalculatorServlet.java

```
package mypack;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
public class CalculatorServlet extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
        out.println("<html><head><title>Servlet CalculatorServlet</title></head><body>");
 double n1 = Double.parseDouble(request.getParameter("txtN1"));
        double n2 = Double.parseDouble(request.getParameter("txtN2"));
        double result =0;
        String opr=request.getParameter("opr");
        if(opr.equals("+")) result=n1+n2;              if(opr.equals("-")) result=n1-n2;
        if(opr.equals("*")) result=n1*n2;              if(opr.equals("/")) result=n1/n2;
        out.println("<h1> Result = "+result);         out.println("</body></html>");}     }
```





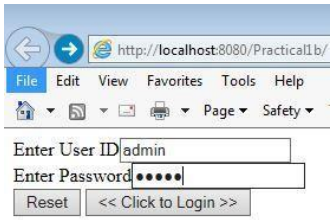
1b. Create a servlet for a login page. If the username and password are correct then it says message "Hello <username>" else a message "login failed"

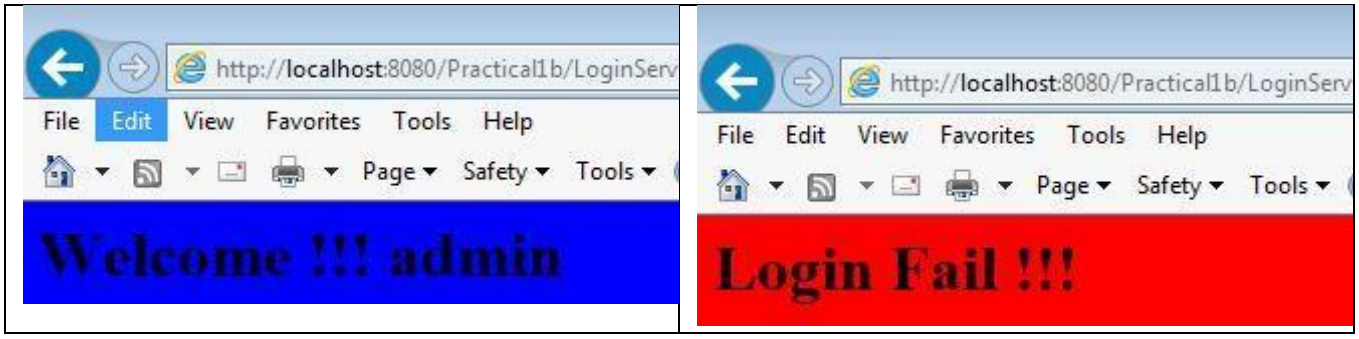
index.html

```
<html><head><title>Login Form</title></head>
<form action="LoginServlet" >
      Enter User ID<input type="text" name="txtId"><br>
      Enter Password<input type="password" name="txtPass"><br>
<input type="reset"><input type="submit" value=" Click to Login " ></form></html>
```

LoginServlet.java

```
package mypack;
import java.io.*;
import javax.servlet.ServletException;
import javax.servlet.http.*;
public class LoginServlet extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
        out.println("<html><head><title>Servlet LoginServlet</title></head>");
        String uname = request.getParameter("txtId");
        String upass = request.getParameter("txtPass");
        if(uname.equals("admin") && upass.equals("12345")){
        out.println("<body bgcolor=blue >");
        out.println("<h1> Welcome !!! "+uname+"</h1>");
        }
        else{
        out.println("<body bgcolor=red >");out.println("<h1> Login Fail !!! </h1>");
        }
        out.println("</body></html>");}}
```





1c. Create a registration servlet in Java using JDBC. Accept the details such as Username, Password, Email, and Country from the user using HTML Form and store the registration details in the database.

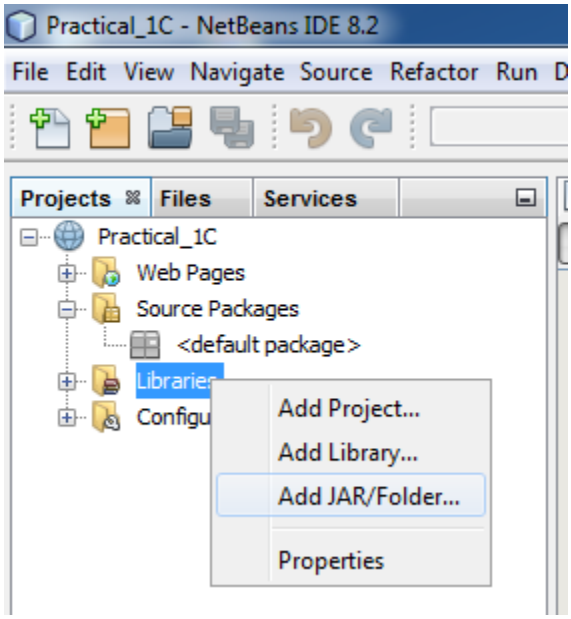
MySql queries

```
create database LoginDB;
use LoginDB;

create table user(username varchar(20) PRIMARY KEY, password varchar(20), email varchar(20), country varchar(20));

insert into user values ('admin','admin','admin@admin.com','India');

select * from user;
```



- Add jar file →**<<Netbean Folder>>**\ide\modules\ext\mysql-connector-java-5.1.23-bin.jar

index.html

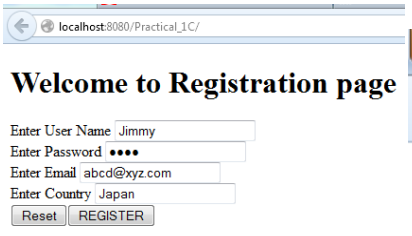
```
<html><head><title>Registration Page</title></head>
<body>
<form action="RegisterServlet" >
<H1>Welcome to Registration page</H1>
      Enter User Name        <input type="text" name="txtUid"><br>
      Enter Password        <input type="password" name="txtPass"><br>
      Enter Email       <input type="text" name="txtEmail" ><br>
      Enter Country       <input type="text" name="txtCon" ><br>
<input type="reset" ><input type="submit" value="REGISTER" >
</form>
</body>
```

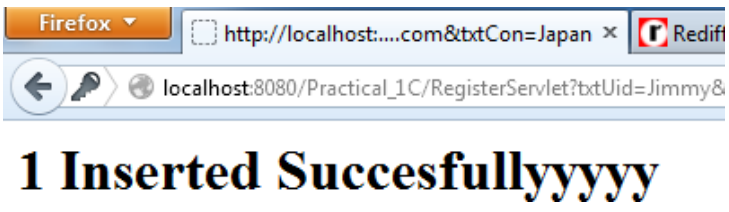
```
</html>
```

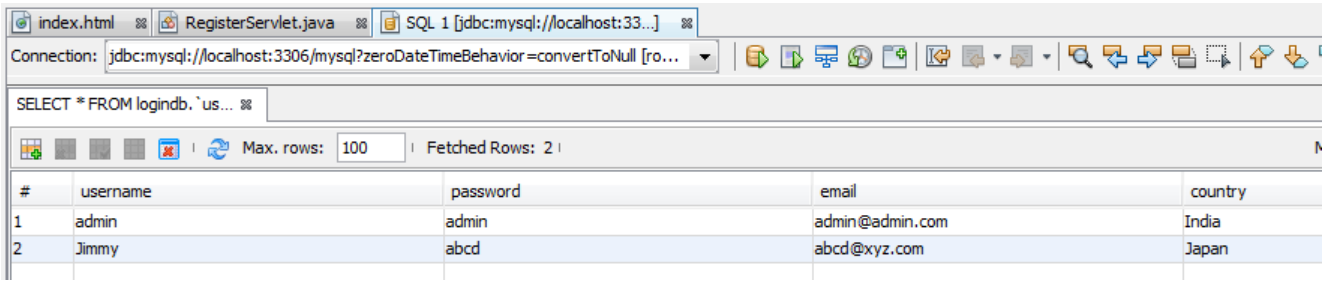
RegisterServlet.java

```
package mypack;
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class RegisterServlet extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
      String id = request.getParameter("txtUid");
      String ps = request.getParameter("txtPass");
      String em = request.getParameter("txtEmail");
      String co = request.getParameter("txtCon");
      try{
      Class.forName("com.mysql.jdbc.Driver");
      Connection con =DriverManager.getConnection("jdbc:mysql://localhost:3306/logindb");
      PreparedStatement pst = con.prepareStatement("insert into user values(?,?,?,?)");
      pst.setString(1,id);
      pst.setString(2,ps);
      pst.setString(3,em);
      pst.setString(4,co);
      int row = pst.executeUpdate();
      out.println("<h1>"+row+ " Inserted Succesfullyyyyy");
      }catch(Exception e){out.println(e);}
}
}
```







2a. Using Request Dispatcher Interface create a Servlet which will validate the password entered by the user, if the user has entered "Servlet" as password, then he will be forwarded to Welcome Servlet else the user will stay on the index.html page and an error message will be displayed.

index.html

```
<html><head><title>Login Form</title></head>
<form action="LoginServlet" >
      Enter User ID<input type="text" name="txtId"><br>
      Enter Password<input type="password" name="txtPass"><br>
<input type="reset">
<input type="submit" value=" Click to Login " >
</form>
</html>
```

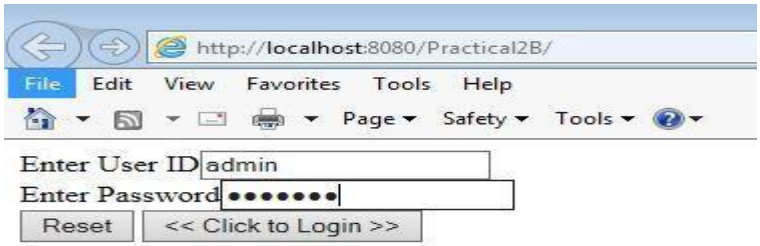
LoginServlet.java

```
package mypack;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.RequestDispatcher;
public class LoginServlet extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
         out.println("<html><head>");
         out.println("<title>Servlet LoginServlet</title></head>");
         String uname = request.getParameter("txtId");
         String upass = request.getParameter("txtPass");
         if(uname.equals("admin") && upass.equals("servlet")){
        RequestDispatcher rd = request.getRequestDispatcher("WelcomeServlet");
        rd.forward(request, response);
         }
         else{
         out.println("<body bgcolor=red >");
         out.println("<h1> Login Fail !!! </h1>");
         RequestDispatcher rd = request.getRequestDispatcher("index.html");
        rd.include(request, response);
         }
         out.println("</body>");
         out.println("</html>");
      }
```

```
    }
```

WelcomeServlet.java

```
package mypack;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.RequestDispatcher;
public class LoginServlet extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
          throws ServletException, IOException {
       response.setContentType("text/html;charset=UTF-8");
       PrintWriter out = response.getWriter();
          out.println("<html><head>");
          out.println("<title>Servlet LoginServlet</title></head>");
          String uname = request.getParameter("txtId");
          String upass = request.getParameter("txtPass");
          if(uname.equals("admin") && upass.equals("servlet")){
         RequestDispatcher rd = request.getRequestDispatcher("WelcomeServlet");
         rd.forward(request, response);
          }
          else{
          out.println("<body bgcolor=red >");
          out.println("<h1> Login Fail !!! </h1>");
          RequestDispatcher rd = request.getRequestDispatcher("index.html");
         rd.include(request, response);
          }
          out.println("</body>");
          out.println("</html>");
       }    }
```







2b. Create a servlet that uses Cookies to store the number of times a user has visited servlet.

index.html

```
<html>
<head><title>Cookie Demo</title></head>
<body>
<form action="Page1" >
        Enter Your Name <input type="text" name="txtName"><br>
<input type="submit" value="~~~ Click to Enter ~~~">
</form>
</body>
</html>
```

Page1.java

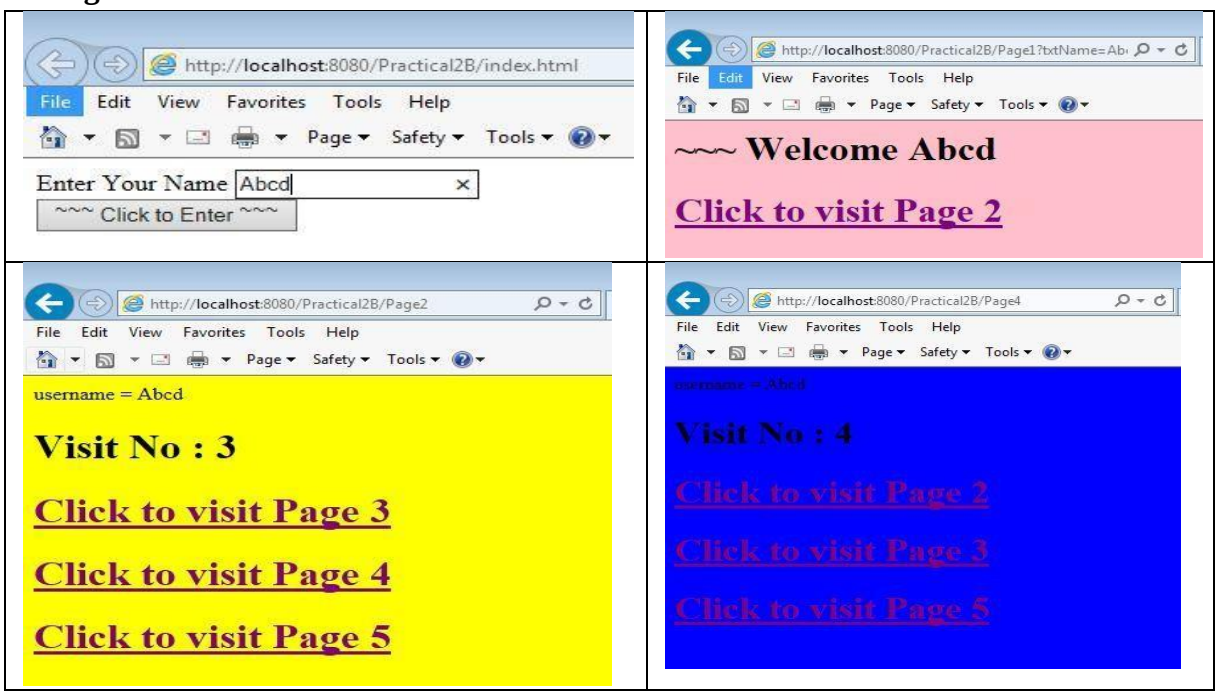
```
package mypack;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.Cookie;
public class Page1 extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
        out.println("<html><head><title>Page1</title></head>");
        out.println("<body bgcolor=pink >");
        String uname = request.getParameter("txtName");
        out.println("<h1>~~~ Welcome "+uname+"</h1>");
        Cookie ck1 = new Cookie("username", uname);
        Cookie ck2 = new Cookie("visit","1");
        response.addCookie(ck1); response.addCookie(ck2);
        out.println("<h1><a href=Page2 >Click to visit Page 2 </a></h1>");
        out.println("</body>");
        out.println("</html>");
      }
   }
```

Page2.java

```
package mypack;
import java.io.*;
import javax.servlet.ServletException;
import javax.servlet.http.*;
public class Page2 extends HttpServlet {
```

```
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
        out.println("<html><head><title>Page2</title></head>");
        out.println("<body bgcolor=yellow >");
        Cookie [] ck = request.getCookies();
        for(int i=0;i<ck.length;i++){
        if(ck[i].getName().equals("visit")){
           int count = Integer.parseInt(ck[i].getValue())+1;
           out.println("<h1>Visit No : "+count+"</h1>");
           ck[i] = new Cookie("visit",count+"");
           response.addCookie(ck[i]);
           }
else {
        out.println(ck[i].getName()+ " = "+ck[i].getValue());          }
        out.println("<h1><a href=Page3 >Click to visit Page 3 </a></h1>");
        out.println("<h1><a href=Page4 >Click to visit Page 4 </a></h1>");
        out.println("<h1><a href=Page5 >Click to visit Page 5 </a></h1>");
        out.println("</body>");
        out.println("</html>");
      }   }
```

Repeat the code from Page2.java for Page3.java, Page4.java and Page5.java with relevant changes.



Important Note: - Run this program on one machine as server and student's machine as client using proxy setting to demonstrate tracking of individual visitor's activity by server using cookie

2c. Create a servlet demonstrating the use of session creation and destruction. Also check whether the user has visited this page first time or has visited earlier also using sessions.

index.html

```
<html>
<head><title>Session Demo</title></head>
<form action="Page1" method="get" >
      Enter User ID <input type="text" name="txtName"><br>
<input type="reset" ><input type="submit" >
</form>
</html>
```

Page1.java

```
package mypack;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
public class Page1 extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
        out.println("<html><head><title>Servlet Page1</title></head>");

        HttpSession hs = request.getSession(true);
        if(hs.isNew()){
        out.println("<body bgcolor=yellow>");
           String name = request.getParameter("txtName");
           hs.setAttribute("uname", name);
           hs.setAttribute("visit", "1");
        out.println("<h1>Welcome First Time</h1>");
        }
        else{
        out.println("<h1>Welcome Again</h1>");
        int visit = Integer.parseInt((String)hs.getAttribute("visit"))+1;
        out.println("<h1>You Visited "+visit+"Times</h1>");
        hs.setAttribute("visit", ""+visit);
        }
        out.println("<h1>Your Session ID "+hs.getId()+"</h1>");
        out.println("<h1>You Logged in at  "+new java.util.Date(hs.getCreationTime())+"</h1>");
        out.println("<h1><a href=Page2>Click for Page 2 </a></h1>");
        out.println("<h1><a href=Page3>Click for Page 3 </a></h1>");
```

```
        out.println("<h1><a href=Page4>Click for Page 4 </a></h1>");
        out.println("<h1><a href=LogoutServlet>Click to Terminate Session </a></h1>");
        out.println("</body>");
        out.println("</html>");
    }
  }
```

Page2.java

```
package mypack;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
public class Page2 extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    PrintWriter out = response.getWriter();
      out.println("<html><head><title>Servlet Page2</title></head>");

        HttpSession hs = request.getSession(false);
        out.println("<h1>Welcome Again on Page No. 2</h1>");
        int visit = Integer.parseInt((String)hs.getAttribute("visit"))+1;
        out.println("<h1>You Visited "+visit+"Times</h1>");
        hs.setAttribute("visit", ""+visit);

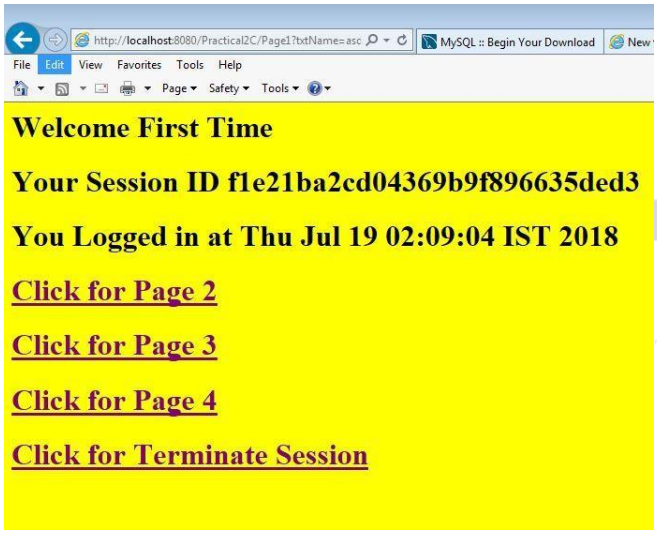
        out.println("<h1>Your Session ID "+hs.getId()+"</h1>");
        out.println("<h1>You Logged in at  "+new java.util.Date(hs.getCreationTime())+"</h1>");
        out.println("<h1><a href=Page1>Click for Page 1 </a></h1>");
        out.println("<h1><a href=Page3>Click for Page 3 </a></h1>");
        out.println("<h1><a href=Page4>Click for Page 4 </a></h1>");
        out.println("<h1><a href=LogoutServlet>Click for Terminate Session </a></h1>");
        out.println("</body>");
        out.println("</html>");
    }
  }
```

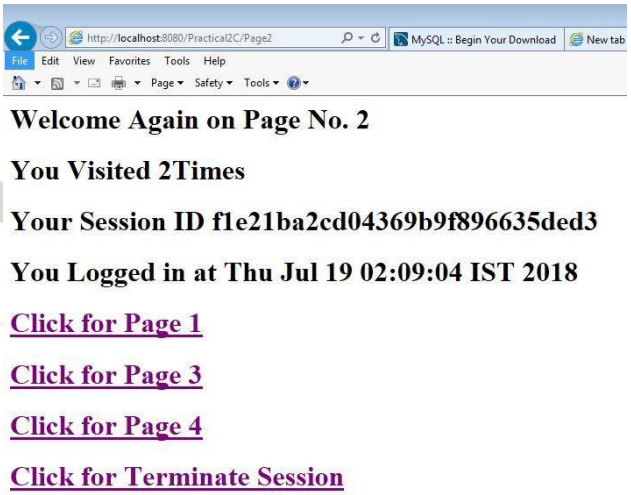
Repeat the code from Page2.java in Page3.java and Page4.java with relevant changes.

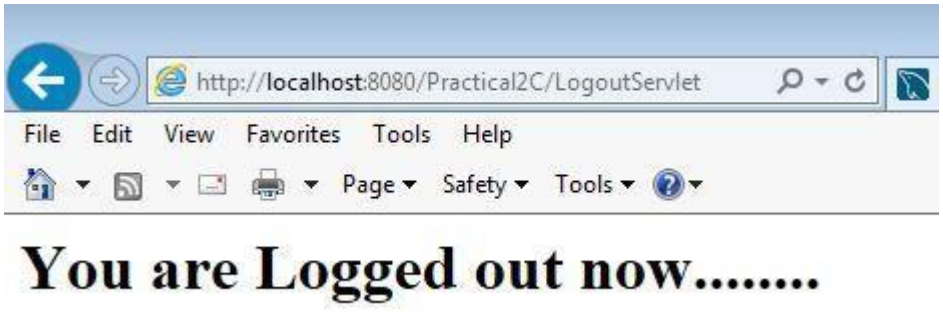
LogoutServlet.java

```
package mypack;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
```

```
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
public class LogoutServlet extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
        out.println("<html><head><title>Servlet LogoutServlet</title></head>");
        out.println("<body>");
        javax.servlet.http.HttpSession hs = request.getSession();
        if(hs != null) hs.invalidate();
        out.println("<h1>You are Logged out now.....</h1>");
        out.println("</body>");
        out.println("</html>");
      }
  }
```







Important Note: - Run this program on one machine as server and student's machine as client using proxy setting to demonstrate tracking of individual visitor's activity by server.

3a. Create a Servlet application to upload and download a file.

~~~~~~~~~~~~~~~~~~~~ **index.html** ~~~~~~~~~~~~~~~~~~~~~~~~

```
<html>
   <body>
   <form action="FileUploadServlet" method="post" enctype="multipart/form-data">
   Select File to Upload:<input type="file" name="file" id="file">
   Destination <input type="text" value="/tmp" name="destination">
   <br>
   <input type="submit" value="Upload file" name="upload" id="upload">
   </form>
   </body>
</html>
```

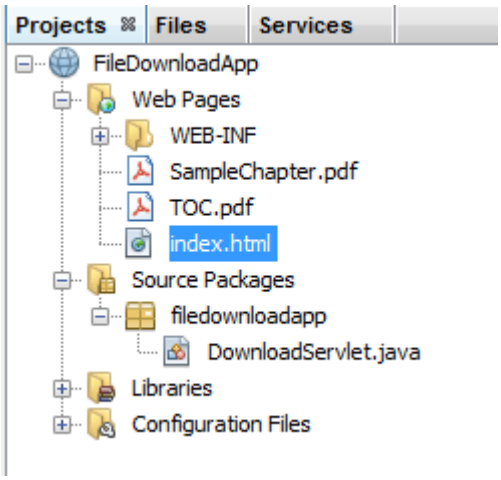
~~~~~ **FileUploadServlet.java** ~~~~~~~

```
package fileservletapp;

import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.MultipartConfig;
import javax.servlet.http.*;

@MultipartConfig
public class FileUploadServlet extends HttpServlet {
public void doPost(HttpServletRequest req,HttpServletResponse res) throws ServletException,
IOException
{
   res.setContentType("text/html");
   PrintWriter out = res.getWriter();
   String path=req.getParameter("destination");
   Part filePart=req.getPart("file");
   String filename=filePart.getSubmittedFileName().toString();
  out.print("<br><br><hr> file name: "+filename);
  OutputStream os=null;
   InputStream is=null;
 try    {
       os=new FileOutputStream(new File(path+File.separator+filename));
       is=filePart.getInputStream();
       int read=0;

       while ((read = is.read()) != -1) {
          os.write(read);
       }
       out.println("<br>file uploaded sucessfully...!!!");
   }
   catch(FileNotFoundException e){out.print(e);}
} }
```



~~~~~ index.html ~~~~~~

```
<html>
   <head>
      <title>File Download Page</title>
   </head>
   <body>
      <h1>File Download Application</h1>
      Click <a href="DownloadServlet?filename=SampleChapter.pdf">Sample Chapter</a>
      <br/><br/>
      Click <a href="DownloadServlet?filename=TOC.pdf">Table Of Contents</a>
   </body>
</html>
```

~~~~~ DownloadServlet.java ~~~~~~

```
package filedownloadapp;

import java.io.IOException;
import  java.io.InputStream;
import java.io.PrintWriter;
import javax.servlet.ServletContext;
import javax.servlet.ServletException;
import javax.servlet.ServletOutputStream;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;



public class DownloadServlet extends HttpServlet {
        public void doGet(HttpServletRequest request, HttpServletResponse response)
```

```
                    throws ServletException, IOException {
        response.setContentType("APPLICATION/OCTET-STREAM");
        String filename = request.getParameter("filename");
        ServletContext context = getServletContext();
            InputStream is = context.getResourceAsStream("/" + filename);
        //ServletOutputStream out = response.getOutputStream(); // any of the two works
        PrintWriter out=response.getWriter();
        response.setHeader("Content-Disposition","attachment; filename=\"" + filename + "\""); //
if comment this statement then it wl ask you about the editor with which you want to open the file

         int i;

            while ((i=is.read()) != -1) {
              out.write(i);
            }
            is.close();
            out.close();
    }

}
```

3b. Develop Simple Servlet Question Answer Application using Database.

```
create database qadb;
use qabd;
create table quiz (qno varchar(5) PRIMARY KEY,question varchar(100), op1 varchar(50), op2
varchar(50), op3 varchar(50), op4 varchar(50), ans varchar(50))
insert into quiz values('001','What is the capital of India??','New
Delhi','Kolkata','Chennai','Mumbai','New Delhi');
insert into quiz values('002','Who was the First President of India??','Dr. Rajendra Prasad','Dr. S.
Radhakrishnan','Ram Nath Kovind','V. V. Giri','Dr. Rajendra Prasad');
insert into quiz values('003','What is ORM','Object Ratio Mean','Object Rotation Measure','Object
Relation Mapping','Oracle Request Management','Object Relation Mapping');
insert into quiz values('004','Unit of Energy is ___','Dozon','Kilo Meter ','Joul','Hertz','Joul')
insert into quiz values('005','   --- is the smallest memory unit.','bit','byte','Kilo Byte','Giga Byte','bit')
```

index.html

```
<html><head><title>Quiz Application</title></head>
<body>
<h1>Welcome to Quiz Servlet </h1>
<h1><a href="QuizServlet" >CLICK TO START QUIZ</a></h1>
</body>
</html>
```

QuizServlet.java

```
package mypack;

import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class QuizServlet extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
out.println("<form action=ShowResult >");
      try {
         Class.forName("com.mysql.jdbc.Driver");
         Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/qadb","root","root");
Statement stmt = con.createStatement();
ResultSet res = stmt.executeQuery("select * from quiz");
out.println("<table border=1 >");
int qno=0;
while(res.next()){
   qno++;
```

```
out.println("<tr><td>"+res.getString(1)+"</td>");
out.println("<td>"+res.getString(2)+"</td></tr>");
out.println("<tr><td><input type=radio name="+qno+"
value="+res.getString(3)+"></td><td>"+res.getString(3)+"</td></tr>");
out.println("<tr><td><input type=radio name="+qno+"
value="+res.getString(4)+"></td><td>"+res.getString(4)+"</td></tr>");
out.println("<tr><td><input type=radio name="+qno+"
value="+res.getString(5)+"></td><td>"+res.getString(5)+"</td></tr>");
out.println("<tr><td><input type=radio name="+qno+"
value="+res.getString(6)+"></td><td>"+res.getString(6)+"</td></tr>");
}
      }catch(Exception e){out.println(e);}
out.println("</table>");
out.println("<input type=reset >");
      out.println("<input type=submit value=SUBMIT >");
       out.println("</form>");       }  }
```

---------- ShowResult.java ----------

```
package mypack;
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class ShowResult extends HttpServlet {
public void doGet(HttpServletRequest request, HttpServletResponse response)
          throws ServletException, IOException {
       response.setContentType("text/html;charset=UTF-8");
       PrintWriter out = response.getWriter();
       try  {
    Class.forName("com.mysql.jdbc.Driver");
  Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/qadb","root","root");
Statement stmt = con.createStatement();
ResultSet res = stmt.executeQuery("select ans from quiz");
int count =0, qno=0;
while(res.next()){
   if(res.getString(1).equals(request.getParameter(""+(++qno))))
   { count++;
   out.println("<h1>Correct </h1>");
   }
   else {
      out.println("<h1>Incorrect </h1>");
   }
}
out.println("<h1>Your Score is "+count+" </h1>");
      }catch(Exception e){out.println(e);}}}
```

select * from quiz

Max. rows: 100 Fetched Rows: 5 Matching Rows:

#		op1	op2	op3	op4	ans
1	tal of India??	New Delhi	Kolkata	Chennai	Mumbai	New Delhi
2	st President of India??	Dr. Rajendra Prasad	Dr. S. Radhakrishnan	Ram Nath Kovind	V. V. Giri	Dr. Rajendra Prasad
3		Object Ratio Mean	Object Rotation Measure	Object Relation Mapping	Oracle Request Management	Object Relation Mapping
4	___	Dozon	Kilo Meter	Joul	Hertz	Joul
5	memory unit.	bit	byte	Kilo Byte	Giga Byte	bit

QuizServlet	ShowResult

Firefox http://localhost:80...tical3B/QuizServlet

localhost:8080/Practical3B/QuizServlet

001 What is the capital of India??
- (●) New Delhi
- ( ) Kolkata
- ( ) Chennai
- ( ) Mumbai

002 Who was the First President of India??
- (●) Dr. Rajendra Prasad
- ( ) Dr. S. Radhakrishnan
- ( ) Ram Nath Kovind
- ( ) V. V. Giri

003 What is ORM
- ( ) Object Ratio Mean
- ( ) Object Rotation Measure
- (●) Object Relation Mapping
- ( ) Oracle Request Management

004 Unit of Energy is ___
- ( ) Dozon
- ( ) Kilo Meter
- (●) Joul
- ( ) Hertz

005 is the smallest memory unit.
- (●) bit
- ( ) byte
- ( ) Kilo Byte
- ( ) Giga Byte

Reset SUBMIT

Firefox http://localh

localhost:8080/Practical3I

Incorrect

Incorrect

Incorrect

Correct

Correct

Your Score is 2

3c. Create simple Servlet application to demonstrate Non-Blocking Read Operation.

```
<html>
<head>
<title>Non Blocking IO</title>
<meta charset="UTF-8">
<meta http-equiv="Refresh" content="0; URL=NonBlockingServlet">
</head>
<body>
</body>
</html>
```

---------- NonBlockingServlrt.java ----------

```
package nonblkapp;

import java.io.*;
import java.net.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class NonBlockingServlet extends HttpServlet {
   protected void service(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      try (PrintWriter out = response.getWriter()) {

         out.println("<h1>FileReader</h1>");
         String filename="/WEB-INF/booklist.txt";
         ServletContext c=getServletContext();
         InputStream in=c.getResourceAsStream(filename);
         String
path="http://"+request.getServerName()+":"+request.getServerPort()+request.getContextPath()+"/Rea
dingNonBloclingServlet";
         URL url=new URL(path);
         HttpURLConnection conn=(HttpURLConnection)url.openConnection();
         conn.setChunkedStreamingMode(2);
         conn.setDoOutput(true);
         conn.connect();
         if(in!=null)
         {
            InputStreamReader inr=new InputStreamReader(in);
            BufferedReader br = new BufferedReader(inr);
            String text="";
            System.out.println("Reading started. ..");
            BufferedWriter bw=new BufferedWriter(new
OutputStreamWriter(conn.getOutputStream()));
            while((text=br.readLine())!=null){
               out.print(text+"<br>");
```

```
                try{
                Thread.sleep(1000);
                out.flush();
                }
                catch(InterruptedException ex){}
}out.print("reading completed....");
            bw.flush();
            bw.close();

        }
      }
   }
}
```

---------- ReadingListener.java ----------

```
package nonblkapp;
import java.io.IOException;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.AsyncContext;
import javax.servlet.ReadListener;
import javax.servlet.ServletInputStream;

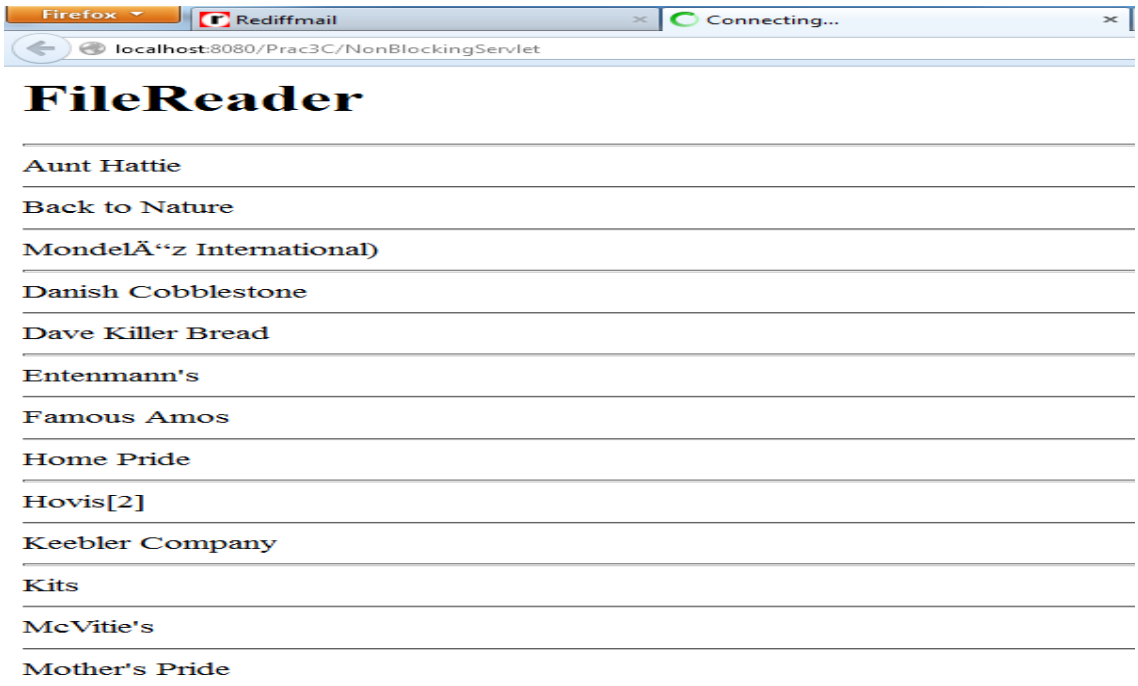
public class ReadingListener implements ReadListener
{
   private ServletInputStream input = null;
   private AsyncContext ac = null;

   ReadingListener(ServletInputStream in, AsyncContext c) {
      input = in;
      ac = c;
   }
   @Override
   public void onDataAvailable() throws IOException {
   }
   public void onAllDataRead() throws IOException {
      ac.complete();
   }
   public void onError(final Throwable t) {
      ac.complete();
      t.printStackTrace();
   }
}
```

---------- ReadingNonBlockingServlet.java ----------

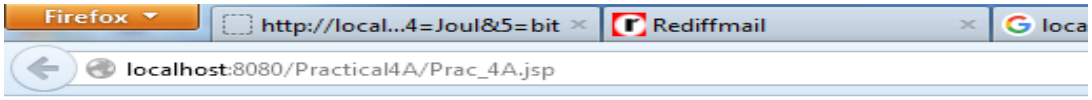
```
package nonblkapp;

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.AsyncContext;
import javax.servlet.ServletException;
import javax.servlet.ServletInputStream;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
/**
 *
 * @author Beena
 */
@WebServlet (name = "ReadingNonBlockingServlet", urlPatterns =
{"/ReadingNonBlockingServlet"},asyncSupported = true )
public class ReadingNonBlockingServlet extends HttpServlet {
@Override
    protected void service(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {
        response.setContentType("text/html");
        AsyncContext ac = request.startAsync();
        ServletInputStream in=request.getInputStream();
        in.setReadListener(new ReadingListener(in,ac));
    }
}
```



4a. Develop a simple JSP application to display values obtained from the use of intrinsic objects of various types.

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<html><head><title>JSP Page</title></head>
<body>
<h1>Use of Intrinsic Objects in JSP</h1>
<h1>Request Object </h1>
      Query String <%=request.getQueryString() %><br>
      Context Path <%=request.getContextPath() %><br>
      Remote Host  <%=request.getRemoteHost()  %><br>
<h1>Response Object </h1>
      Character Encoding Type <%=response.getCharacterEncoding() %><br>
      Content Type       <%=response.getContentType()  %><br>
      Locale             <%=response.getLocale()  %><br>
<h1>Session Object </h1>
      ID <%=session.getId() %><br>
      Creation Time <%=new java.util.Date(session.getCreationTime()) %><br>
      Last Access Time<%=new java.util.Date(session.getLastAccessedTime())  %><br>
</body>
</html>
```



Use of Intrinsic Objects in JSP

Request Object

Query String null
Context Path /Practical4A
Remote Host 0:0:0:0:0:0:0:1

Response Object

Character Encoding Type UTF-8
Content Type text/html;charset=UTF-8
Locale en_US

Session Object

ID 39a6dd44e151027609e21c8b80c6
Creation Time Thu Jul 19 23:03:19 IST 2018
Last Access TimeThu Jul 19 23:09:20 IST 2018

4b. Develop a simple JSP application to pass values from one page to another with validations. (Name-txt, age-txt, hobbies-checkbox, email-txt, gender-radio button).

index.html

```
<html><head><title>User Information Paage</title>
</head>
<body>
<form action="Validate.jsp">
        Enter Your Name<input type="text" name="name" ><br>
        Enter Your Age<input type="text" name="age" ><br>
        Select Hobbies
<input type="checkbox" name="hob" value="Singing">Singing
<input type="checkbox" name="hob" value="Reading">Reading Books
<input type="checkbox" name="hob" value="Football">Playing Football<br>
        Enter E-mail<input type="text" name="email" ><br>
        Select Gender
<input type="radio" name="gender" value="male">Male
<input type="radio" name="gender" value="female">Female
<input type="radio" name="gender" value="other">Other<br>
<input type="hidden" name="error" value="">
<input type="submit" value="Submit Form">
</form>
</body>
</html>
```

---------- Validate.jsp----------

```
<%@page contentType="text/html" pageEncoding="UTF-8" import="mypack.*" %>
<html><head><title>JSP Page</title></head>
<body>
<h1>Validation Page</h1>

<jsp:useBean id="obj"  scope="request"
      class="mypack.CheckerBean"      >
<jsp:setProperty name="obj" property="*"/>
</jsp:useBean>

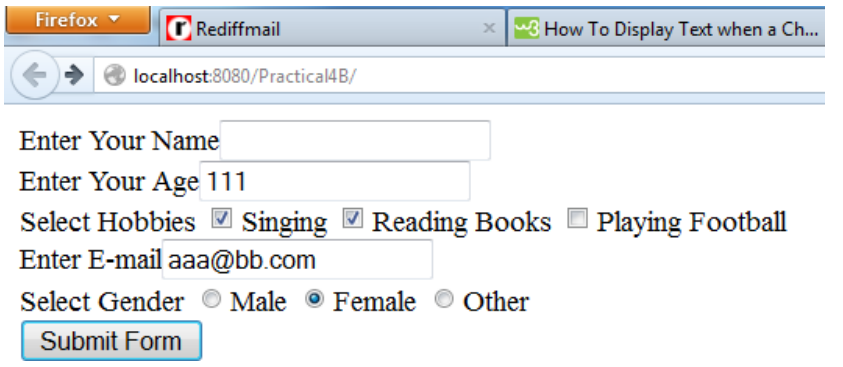
<%if (obj.validate())
{ %>
<jsp:forward page="successful.jsp"/>
<%   }
else  {%>
<jsp:include page="index.html"/>
<%}%>
<%=obj.getError() %>
</body></html>
```

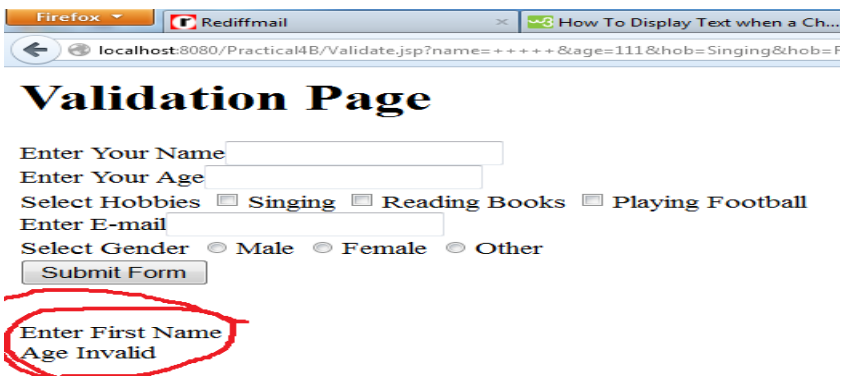
----------CheckerBean.java----------

```
package mypack;

public class CheckerBean {
private String name, age, hob, email, gender, error;
public CheckerBean(){error="";}
public void setName(String n){name=n;}
public void setAge(String a){age=a;}
public void setHob(String h){hob=h;}
public void setEmail(String e){email=e;}
public void setGender(String g){gender=g;}
public void setError(String e){error=e;}
public String getName(){return name;}
public String getAge(){return age;}
public String getHob(){return hob;}
public String getEmail(){return email;}
public String getGender(){return gender;}
public String getError(){return error;}
public boolean validate(){
boolean res=true;
if(name.trim().equals("")) {error+="<br>Enter First Name";res=false;}
if(age.length() > 2 )
{error+="<br>Age Invalid";res=false;}

return res;
}
}
```





4c. Create a registration and login JSP application to register and authenticate the user based on username and password using JDBC.

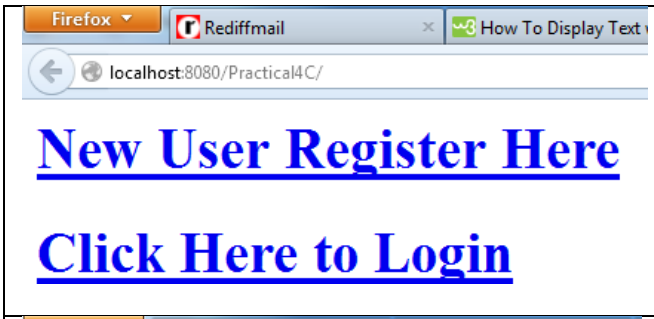
----------Register.html----------

```
<html><head><title>New User Registration Page</title></head>
<body>
<form action="Register.jsp" >
<h1> New User Registration Page</h1>
        Enter User Name <input type="text" name="txtName" ><br>
        Enter Password <input type="password" name="txtPass1" ><br>
        Re-Enter Password<input type="password" name="txtPass2" ><br>
        Enter Email<input type="text" name="txtEmail" ><br>
        Enter Country Name <input type="text" name="txtCon" ><br>
<input type="reset" ><input type="submit" value="REGISTER" >
</form>
</body>
</html>
```

----------Register.jsp----------

```
<%@page contentType="text/html" import="java.sql.*"%>
<html><body>
<h1>Registration JSP Page</h1>
<%
      String uname=request.getParameter("txtName");
      String pass1 = request.getParameter("txtPass1");
      String pass2 = request.getParameter("txtPass2");
      String email = request.getParameter("txtEmail");
      String ctry = request.getParameter("txtCon");
 if(pass1.equals(pass2)){
      try{
  Class.forName("com.mysql.jdbc.Driver");
  Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/logindb");
  PreparedStatement stmt = con.prepareStatement("insert into user values (?,?,?,?)");
  stmt.setString(1, uname);    stmt.setString(2, pass1);
  stmt.setString(3, email);    stmt.setString(4, ctry);
  int row = stmt.executeUpdate();
  if(row==1) { out.println("Registration Successful"); }
  else {
     out.println("Registration FFFFFAAAIIILLLL !!!!");
     %><jsp:include page="Register.html" ></jsp:include>
<%
      }
  }catch(Exception e){out.println(e);}
 }
 else
 {
 out.println("<h1>Password Mismatch</h1>");
```

```
 %>
<jsp:include page="Register.html" ></jsp:include>
<%  }
   %>
</body>
</html>
```











---------- Login.html ----------

```
<html>
<body>
<h1>Login Page</h1>
<form action="Login.jsp" >
        Enter User Name <input type="text" name="txtName" ><br>
        Enter Password <input type="password" name="txtPass" ><br>
<input type="reset" ><input type="submit" value="~~~LOGIN~~" >
</form>
</body>
</html>
```

---------- Login.jsp----------

```
<%@page contentType="text/html" import="java.sql.*"%>
<html><body>
<h1>Registration JSP Page</h1>
<%
        String uname=request.getParameter("txtName");
        String pass = request.getParameter("txtPass");
             try{
    Class.forName("com.mysql.jdbc.Driver");
    Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/logindb");
    PreparedStatement stmt = con.prepareStatement("select password from user where username=?");
    stmt.setString(1, uname);
    ResultSet rs = stmt.executeQuery();
    if(rs.next()){
        if(pass.equals(rs.getString(1)))
        {
           out.println("<h1>~~~ LOGIN SUCCESSFULLL ~~~ </h1>");
        }
        }
    else{
    out.println("<h1>User Name not exist !!!!!</h1>");
    %>
<jsp:include page="Register.html" ></jsp:include>
<%
    }
        }catch(Exception e){out.println(e);}
  %>
</body></html>
```

5a. Create an html page with fields, eno, name, age, desg, salary. Now on submit this data to a JSP page which will update the employee table of database with matching eno.

```
create table emp(empid varchar(10) PRIMARY KEY, ename varchar(50), salary varchar(50),age varchar(50) )
insert into emp values('1','aaa','221234','11')
insert into emp values('2','bbb','334567','22')
insert into emp values('3','ccc','44454','33')
insert into emp values('4','ddd','55123','44')
---- index.html ---------
<html>
<body>
<form action="UpdateEmp.jsp" >
        Enter Employee Number<input type="text" name="txtEno" ><br>
        Enter Name<input type="text" name="txtName" ><br>
        Enter age<input type="text" name="txtAge" ><br>
        Enter Salary<input type="text" name="txtSal" ><br>
<input type="reset" ><input type="submit">

</form>
</body>
</html>
- - - - - - - - - - - - - UpdateEmp.java - - - - - - - - - - - - - - - - - - - -
<%@page contentType="text/html" import="java.sql.*" %>
<html><body>
<h1>Employee Record Update</h1>
<%
      String eno=request.getParameter("txtEno");
      String name=request.getParameter("txtName");
      String age = request.getParameter("txtAge");
      String sal = request.getParameter("txtSal");
           try{
   Class.forName("com.mysql.jdbc.Driver");
   Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/empdb");
   PreparedStatement stmt = con.prepareStatement("select * from emp where empid=?");
   stmt.setString(1, eno);
   ResultSet rs = stmt.executeQuery();
   if(rs.next()){
      out.println("<h1>~~~ Employee "+name+" Exist ~~~ </h1>");
      PreparedStatement pst1= con.prepareStatement("update emp set salary=? where empid=?");
      PreparedStatement pst2= con.prepareStatement("update emp set age=? where empid=?");
      pst1.setString(1, sal); pst1.setString(2, eno);
      pst2.setString(1, age); pst2.setString(2, eno);
      pst1.executeUpdate();       pst2.executeUpdate();
   }
```

```
else{
out.println("<h1>Employee Record not exist !!!!!</h1>");

}
   }catch(Exception e){out.println(e);}
%></body></html>
```

SELECT * FROM emp LIMIT 1... ⊠

Max. rows: 100 | Fetched Rows: 5

#	empid	ename	salary	age
1	1	aaaa	12345	11
2	2	bbbb	2222	22
3	3	cccc	12121	33
4	4	ddddd	123123	44
5	5	eeeee	12123	55

SELECT * FROM emp LIMIT 1... ⊠

Max. rows: 100 | Fetched Rows: 5

#	empid	ename	salary	age
1	1	aaaa	54321	99
2	2	bbbb	2222	22
3	3	cccc	12121	33
4	4	ddddd	123123	44

5b. Create a JSP page to demonstrate the use of Expression language.

<< any example to demonstrate the use of EL like calculator or any formulas etc…. >>

5c. Create a JSP application to demonstrate the use of JSTL.

Basic insert, update and delete example using core and sql tag libraries in JSTL.

```
CREATEDATABASEIF NOTEXISTS sampleDB;
CREATETABLE`product` (
  `id` int(10) unsigned NOTNULLauto_increment,
  `pname` varchar(45) NOTNULL,
  `quantity` int(10) unsigned NOTNULL,
  PRIMARYKEY (`id`)
);

INSERTINTO`product` (`id`,`pname`,`quantity`) VALUES
 (1,'Mouse',50),
 (2,'Keyboard',5),
 (3,'Monitor',34);
```

1. index.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <metahttp-equiv="Content-Type"content="text/html; charset=UTF-8">
        <title>JSP Page</title>
    </head>
    <body>
        <h1>Choose Option</h1>
        <ahref="insert.jsp">Insert Record</a><p></p>
        <ahref="display.jsp">Display Record</a>
    </body>
</html>
```

2. insert.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<!DOCTYPE html>
<html>
    <head>
        <metahttp-equiv="Content-Type"content="text/html; charset=UTF-8">
        <title>JSP Page</title>
    </head>
    <body>
        <formaction="insertdb.jsp"method="post">
            <tableborder="0"cellspacing="2"cellpadding="5">
                <thead>
                    <tr>
                        <thcolspan="2">Purchase Product</th>
                    </tr>
                </thead>
                <tbody>
                    <tr>
                        <td><label>Product Name</label></td>
                        <td><inputtype="text"name="pname"/></td>
                    </tr>
                    <tr>
                        <td><label>Quantity</label></td>
                        <td><inputtype="text"name="qty"/></td>
                    </tr>
                    <tr>
                        <td><inputtype="submit"value="Save"/></td>
                        <td><inputtype="reset"value="reset"/></td>
                    </tr>
```

```
                </tbody>
            </table>
        </form>
        <fontcolor="red"><c:iftest="${not empty param.errMsg}">
            <c:outvalue="${param.errMsg}"/>
            <ahref="index.jsp">Go Back</a>
        </c:if></font>
        <fontcolor="green"><c:iftest="${not empty param.susMsg}">
            <c:outvalue="${param.susMsg}"/>
            <ahref="index.jsp">Go Back</a>
        </c:if></font>

    </body>
</html>
```

3. insertdb.jsp

```
<%@ page import="java.io.*,java.util.*,java.sql.*"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql"%>

<html>
    <head>
        <title>JINSERT Operation</title>
    </head>
    <body>
        <c:iftest="${ empty param.pname or empty param.qty}">
            <c:redirecturl="insert.jsp">
                <c:paramname="errMsg"value="Please Enter Product and Quantity"/>
            </c:redirect>

        </c:if>
        <sql:setDataSourcevar="dbsource"driver="com.mysql.jdbc.Driver"
                        url="jdbc:mysql://localhost/sampleDB"
                        user="root" password="pass"/>


        <sql:updatedataSource="${dbsource}"var="result">
            INSERT INTO product(pname, quantity) VALUES (?,?);
            <sql:paramvalue="${param.pname}"/>
            <sql:paramvalue="${param.qty}"/>
        </sql:update>
        <c:iftest="${result>=1}">
            <fontsize="5"color='green'> Congratulations ! Data inserted
            successfully.</font>

            <c:redirecturl="insert.jsp">
                <c:paramname="susMsg"value="Congratulations ! Data inserted
            successfully." />
            </c:redirect>
        </c:if>


    </body>
</html>
```

4. display.jsp

```
<%@ page import="java.io.*,java.util.*,java.sql.*"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql"%>

<html>
    <head>
```

```
        <title>SELECT Operation</title>
        <script>
            function confirmGo(m,u) {
                if ( confirm(m) ) {
                    window.location = u;
                }
            }
        </script>
    </head>
    <body>

        <sql:setDataSourcevar="dbsource"driver="com.mysql.jdbc.Driver"
                           url="jdbc:mysql://localhost/sampleDB"
                           user="root" password="pass"/>

        <sql:querydataSource="${dbsource}"var="result">
            SELECT * from product;
        </sql:query>
    <center>
        <form>
            <tableborder="1"width="40%">
                <caption>Product List</caption>
                <tr>
                    <th>Product ID</th>
                    <th>Product Name</th>
                    <th>Quantity</th>
                    <thcolspan="2">Action</th>
                </tr>
                <c:forEachvar="row"items="${result.rows}">
                    <tr>
                        <td><c:outvalue="${row.id}"/></td>
                        <td><c:outvalue="${row.pname}"/></td>
                        <td><c:outvalue="${row.quantity}"/></td>
                        <td><ahref="update.jsp?id=<c:out
value="${row.id}"/>">Update</a></td>
                        <td><ahref="javascript:confirmGo('Sure to delete this
record?','deletedb.jsp?id=<c:out value="${row.id}"/>')">Delete</a></td>

                    </tr>
                </c:forEach>
            </table>
        </form>
        <ahref="index.jsp">Go Home</a>
    </center>
</body>
</html>
```

5. update.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@ page import="java.io.*,java.util.*,java.sql.*"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql"%>
<!DOCTYPE html>
<html>
    <head>
        <metahttp-equiv="Content-Type"content="text/html;  charset=UTF-8">
        <title>JSP Page</title>
    </head>
    <body>

        <sql:setDataSourcevar="dbsource"driver="com.mysql.jdbc.Driver"
                           url="jdbc:mysql://localhost/sampleDB"
```

```
                        user="root" password="pass"/>

        <sql:querydataSource="${dbsource}"var="result">
            SELECT * from product where id=?;
            <sql:paramvalue="${param.id}"/>
        </sql:query>
        <formaction="updatedb.jsp"method="post">
            <tableborder="0"width="40%">
                <caption>Update Product</caption>
                <tr>
                    <th>Product Name</th>
                    <th>Quantity</th>
                </tr>
                <c:forEachvar="row"items="${result.rows}">
                    <tr>
                        <td><inputtype="hidden"value="${param.id}"name="id"/>
                            <inputtype="text"value="${row.pname}"name="pname"/></td>
                      <td><inputtype="text"value="${row.quantity}"name="qty"/></td>
                        <td><inputtype="submit"value="Update"/></td>
                    </tr>
                </c:forEach>
            </table>
            <ahref="index.jsp">Go Home</a>
        </form>
    </body>
</html>
```

6. updatedb.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@ page import="java.io.*,java.util.*,java.sql.*"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql"%>
<!DOCTYPE html>
<html>
    <head>
        <metahttp-equiv="Content-Type"content="text/html;  charset=UTF-8">
        <title>JSP Page</title>
    </head>
    <body>
        <sql:setDataSourcevar="dbsource"driver="com.mysql.jdbc.Driver"
                        url="jdbc:mysql://localhost/sampleDB"
                        user="root" password="pass"/>
        <sql:updatedataSource="${dbsource}"var="count">
            UPDATE product SET pname = ?, quantity=?
            WHERE id='${param.id}'
            <sql:paramvalue="${param.pname}"/>
            <sql:paramvalue="${param.qty}"/>
        </sql:update>
        <c:iftest="${count>=1}">
            <fontsize="5"color='green'> Congratulations ! Data updated
            successfully.</font>
              <ahref="index.jsp">Go Home</a>
        </c:if>
    </body>
</html>
```

7. deletedb.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@ page import="java.io.*,java.util.*,java.sql.*"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/sql" prefix="sql"%>
<!DOCTYPE html>
<html>
```

```
    <head>
        <metahttp-equiv="Content-Type"content="text/html;  charset=UTF-8">
        <title>JSP Page</title>
    </head>
    <body>
        <sql:setDataSourcevar="dbsource"driver="com.mysql.jdbc.Driver"
                          url="jdbc:mysql://localhost/sampleDB"
                          user="root" password="pass"/>
        <sql:updatedataSource="${dbsource}"var="count">
            DELETE FROM product
            WHERE id='${param.id}'
        </sql:update>
        <c:iftest="${count>=1}">
            <fontsize="5"color='green'> Congratulations ! Data deleted
            successfully.</font>
              <ahref="index.jsp">Go Home</a>
        </c:if>
    </body>
</html>
```

6a. Create a Currency Converter application using EJB.

---------- **index.html** ----------

```
<html><head><title>Currency Converter</title></head>
<body>
<form action="CCServlet" >
        Enter Amount <input type="text" name="amt"><br>
        Select Conversion Type
<input type="radio" name="type" value="r2d" checked>Rupees to Dollar
<input type="radio" name="type" value="d2r" >Dollor to Rupees<br>
<input type="reset" ><input type="submit" value="CONVERT" >
</form>
</body>
</html>
```

---------- **CCServlet.java** ----------

```
package mypack;
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.ejb.EJB;
import mybeans.CCBean;
public class CCServlet extends HttpServlet {
   @EJB CCBean obj;
public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
PrintWriter out = response.getWriter();
double amt = Double.parseDouble(request.getParameter("amt"));
if(request.getParameter("type").equals("r2d"))
{
out.println("<h1>"+amt+ " Rupees = "+obj.r2Dollor(amt)+" Dollors</h1>");
}
if(request.getParameter("type").equals("d2r"))
{
out.println("<h1>"+amt+ " Dollors = "+obj.d2Rupees(amt)+" Rupees</h1>");
}

}}
```

---------- CCBean ----------

Create a stateless session bean

```
package mybeans;

import javax.ejb.Stateless;

@Stateless
```

```
public class CCBean {
public CCBean(){}
public double r2Dollor(double r){   return r/65.65;  }
public double d2Rupees(double d){   return d*65.65;  }
}
```





6b. Develop a Simple Room Reservation System Application Using EJB.

Create table rookbook(RoomId varchar(4) PRIMARY KEY, RoomType varchar(20), charges number(5,2), cust varchar(20), mob varchar(20) , status varchar(10))
insert into roombook values('1001','Delux',5000.00,",",'Not Booked')
insert into roombook values('1002','Super Delux',7000.00,",",'Not Booked')
insert into roombook values('1003','Suit',9500.00,",",'Not Booked')
insert into roombook values('2001','Delux',5000.00,",",'Not Booked')
insert into roombook values('2002','Super Delux',7000.00,",",'Not Booked')
insert into roombook values('2003','Suit',9500.00,",",'Not Booked')

----------RoomBook.html ----------

```
<form action="RBServlet" >
   Select a room Type
<input type="radio" name="txtType" value="Delux">Delux
<input type="radio" name="txtType" value="Super Delux">Super Delux
<input type="radio" name="txtType" value="Suit">Suit<br>
   Enter Your Name<input type="text" name="txtCust" ><br>
   Enter Mobile No.<input type="text" name="txtMob" ><br>
<input type="reset" ><input type="submit" value="Book Room">
</form>
```

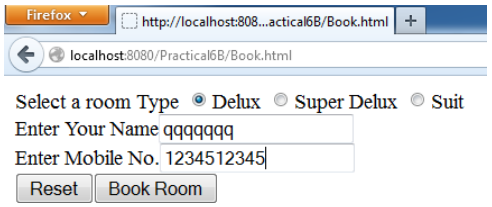
---------- RBServlet----------

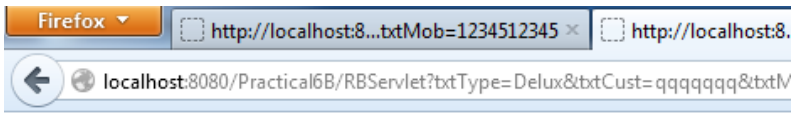
```
package mypack;
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.ejb.EJB;
import mybeans.RRBean;
public class RBServlet extends HttpServlet {
@EJB RRBean obj;
public void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException{
PrintWriter out=response.getWriter();
String rt=request.getParameter("txtType");
String cn=request.getParameter("txtCust");
String cm=request.getParameter("txtMob");
String msg = obj.roomBook(rt, cn, cm);
      out.println(msg);
}}
```

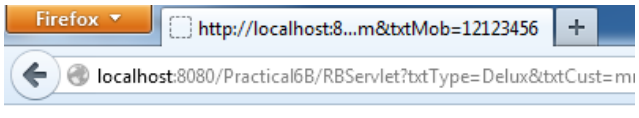
----------RRBean.java ----------

```
package mybeans;
import javax.ejb.Stateless;
import java.sql.*;
@Stateless
public class RRBean {
```

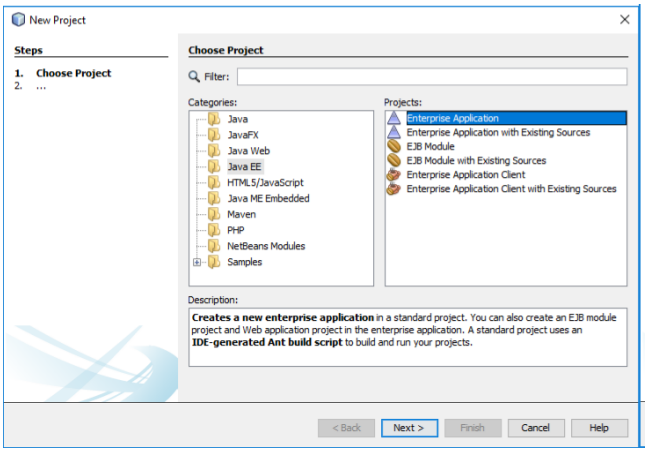
```
public RRBean(){}
public String roomBook(String rt, String cn, String cm){
String msg="";
try{
Class.forName("com.mysql.jdbc.Driver");
Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/rrdb","root","root");
String query="select * from roombook where RoomType=? and status='Not Booked'";
PreparedStatement pst = con.prepareStatement(query);
pst.setString(1,rt);
ResultSet rs= pst.executeQuery();
if(rs.next()){
   String rno=rs.getString(1);
PreparedStatement stm1 = con.prepareStatement("update roombook set cust=? where RoomId=? ");
PreparedStatement stm2 = con.prepareStatement("update roombook set mobile=? where RoomId=? ");
PreparedStatement stm3 = con.prepareStatement("update roombook set status=? where RoomId=? ");
stm1.setString(1,cn);        stm1.setString(2,rno);
stm2.setString(1,cm);         stm2.setString(2,rno);
stm3.setString(1, "Booked"); stm3.setString(2,rno);
stm1.executeUpdate();
stm2.executeUpdate();
stm3.executeUpdate();
msg = "Room "+rno+ " Booked <br> Charges = "+rs.getString(3);
}
else
{
msg = "Room "+rt+ " currently Not available";
}
}catch(Exception e){msg=""+e;}
return msg;}}
```

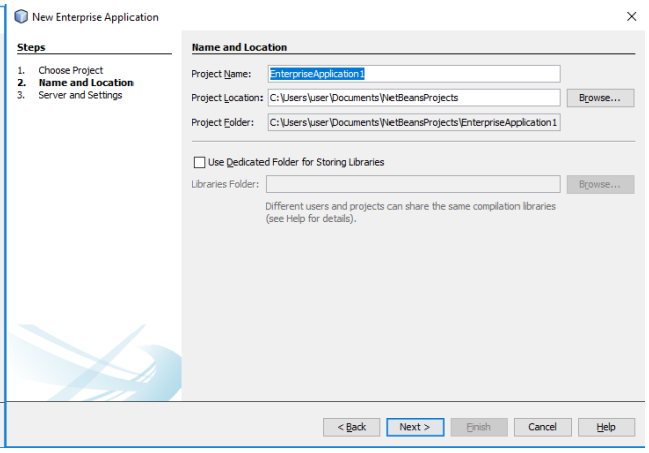


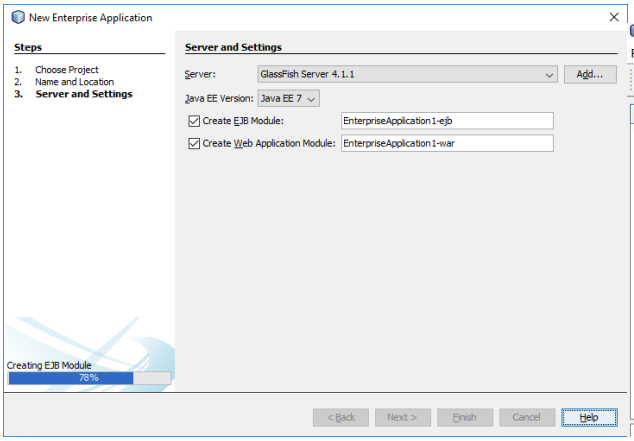


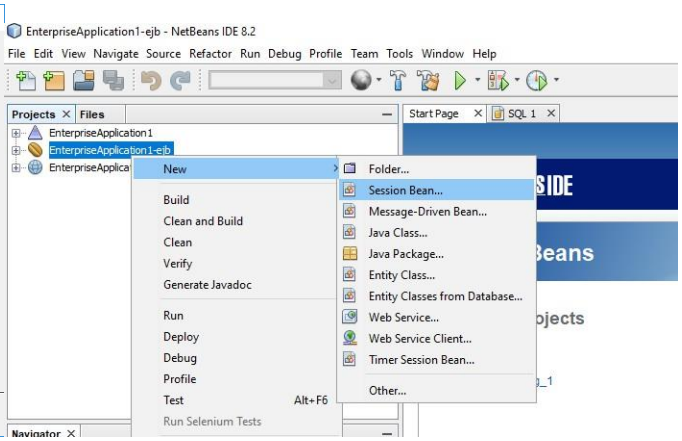


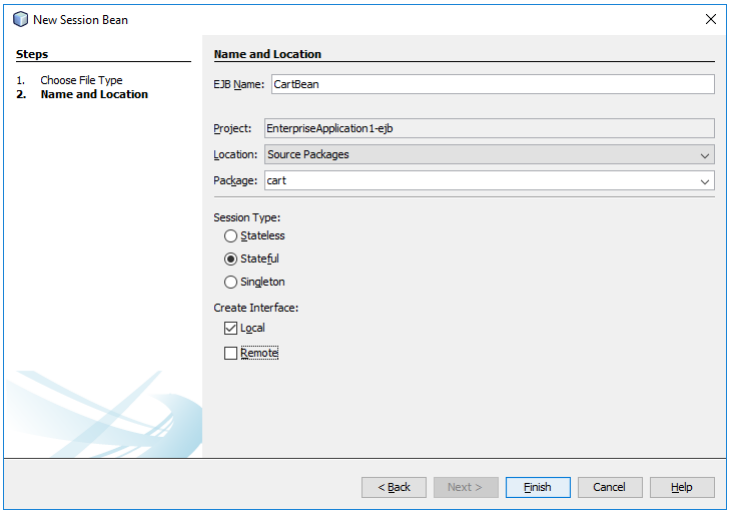
6c. Develop simple shopping cart application using EJB [Stateful Session Bean].











CartBeanLocal.java

```
package cart;

import java.util.List;
import javax.ejb.Local;

@Local
public interface CartBeanLocal {
   public void initialize(String person) throws Exception;
   public void initialize(String person, String id)
       throws Exception;
   public void addBook(String title);
   public void removeBook(String title) throws Exception;
   public List<String> getContents();
   public void remove();

}
```

CartBean.java

~~~~~~~~~~~~~

```
package cart;

import java.util.ArrayList;
import java.util.List;
import javax.ejb.Remove;
import javax.ejb.Stateful;


@Stateful
public class CartBean implements CartBeanLocal {
   String customerName;
   String customerId;
   List<String> contents;

   public void initialize(String person, String id)
             throws Exception {
      if (person == null) {
         throw new Exception("Null person not allowed.");
      } else {

         customerName = person;
      }

      if ( person=="ABC" && id=="123") {
```
~~~~~~~~~~~~~

```
            customerId = id;
        } else {
            throw new Exception("Invalid id: " + id);
        }

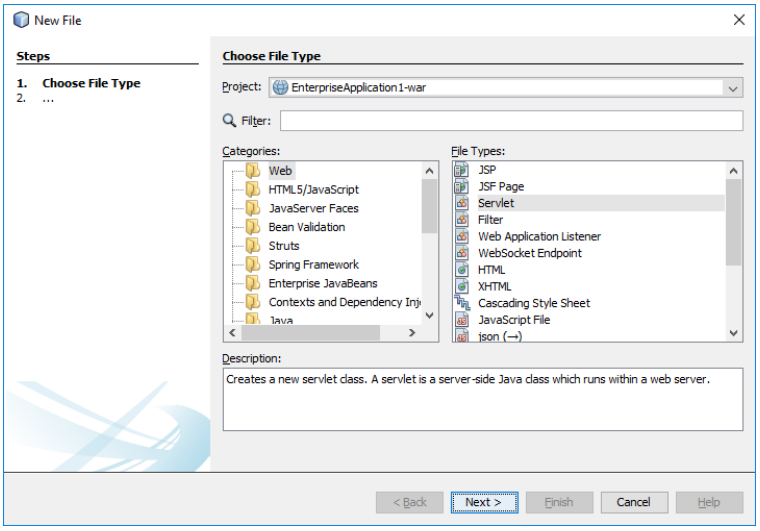
        contents = new ArrayList<String>();
    }

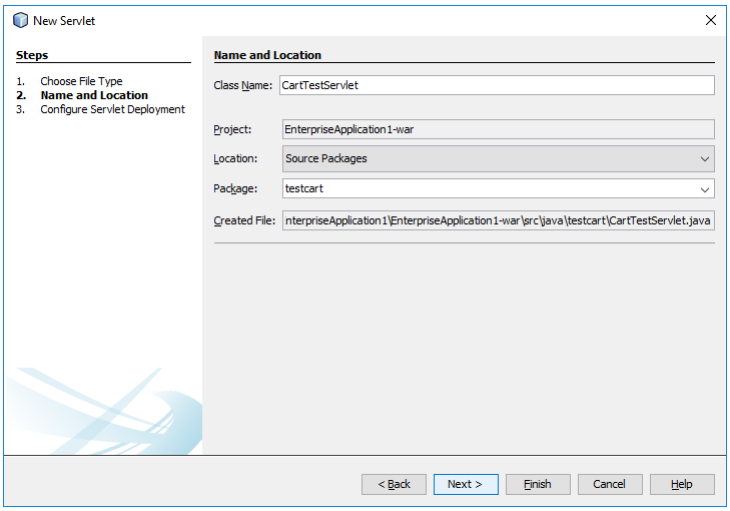
    public void addBook(String title) {
        contents.add(title);
    }

    public void removeBook(String title) throws Exception {
        boolean result = contents.remove(title);
        if (result == false) {
            throw new Exception(title + " not in cart.");
        }
    }

    public List<String> getContents() {
        return contents;
    }

    @Remove
    public void remove() {
        contents = null;
    }
}
```






```
package testcart;

import cart.CartBeanLocal;
import java.io.*;
import java.util.*;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.naming.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

@WebServlet(name = "CartTestServlet", urlPatterns = {"/CartTestServlet"})
public class CartTestServlet extends HttpServlet {

    CartBeanLocal cartBean = lookupCartBeanLocal();

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try{
            cartBean.initialize("ABC", "123");
        }catch(Exception e){}
        cartBean.addBook("Java 8 Cookbook");
        cartBean.addBook("Enterprise Java 7 ");
        cartBean.addBook("Java for Dummies");
        cartBean.addBook("Learn Java 8");

        try (PrintWriter out = response.getWriter()) {
```

```
            try{

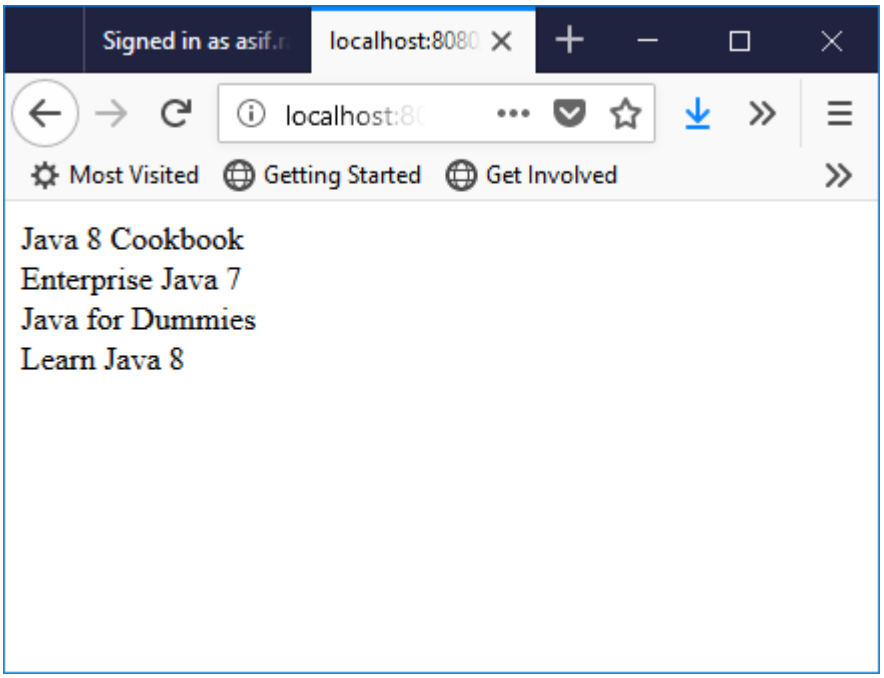
                List<String> books = cartBean.getContents();

                for( String s : books)
                    out.println(s +"<br />");

            }catch(Exception e){}
        }
    }

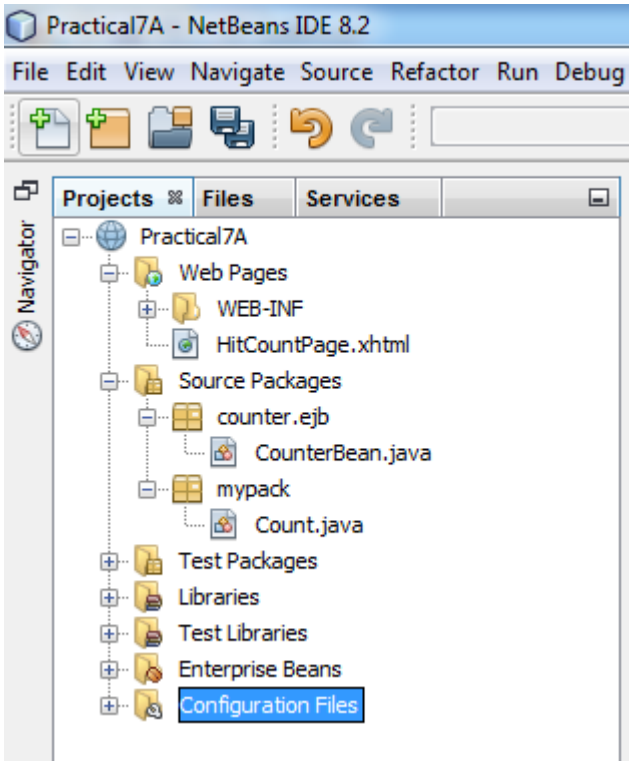
    private CartBeanLocal lookupCartBeanLocal() {
        try {
            Context c = new InitialContext();
            return (CartBeanLocal) c.lookup("java:global/EnterpriseApplication1/EnterpriseApplication1-
ejb/CartBean!cart.CartBeanLocal");
        } catch (NamingException ne) {
            Logger.getLogger(getClass().getName()).log(Level.SEVERE, "exception caught", ne);
            throw new RuntimeException(ne);
        }
    }

}
```



7a. Develop simple EJB application to demonstrate Servlet Hit count using Singleton Session Beans.

Organization of file in Project Folder



---------- **CounterBean.java** ----------

```
package counter.ejb;
import javax.ejb.Singleton;

@Singleton
public class CounterBean {
   private int hits = 1;

   // Increment and return the number of hits
   public int getHits() {
      return hits++;
   }
}
```

---------- Count.java ----------

```
package mypack;
import java.io.Serializable;
import javax.ejb.EJB;
import javax.enterprise.context.ConversationScoped;
import javax.inject.Named;
import counter.ejb.CounterBean;
@Named("count")
@ConversationScoped
public class Count implements Serializable {
   @EJB
```

```
    private CounterBean counterBean;

    private int hitCount;

    public Count() {
        this.hitCount = 0;
    }

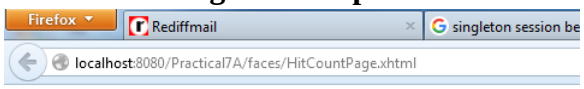
    public int getHitCount() {
        hitCount = counterBean.getHits();
        return hitCount;
    }

    public void setHitCount(int newHits) {
        this.hitCount = newHits;
    }
}
```

---------- HitCountPage.xhtml ----------

```
<?xml version='1.0' encoding='UTF-8' ?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml"
      xmlns:h="http://xmlns.jcp.org/jsf/html">
<h:head>
<title>Page Hit Counter Using Singleton Session Bean ~~~~ </title>
</h:head>
<h:body>
<h1>Welcome to Hit Count Page</h1>
       Page was hit  #{count.hitCount}   times
</h:body>
</html>
```

Run this file to get the output



Welcome to Hit Count Page

Page was hit 4 times

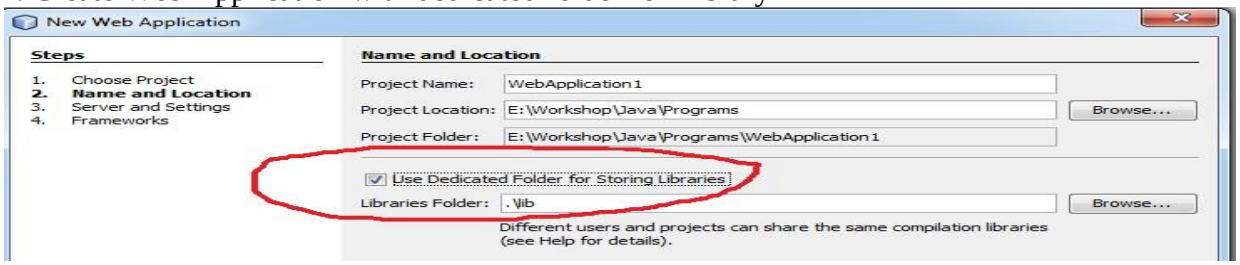
7b. Develop simple visitor Statistics application using Message Driven Bean [Stateless Session Bean].
7c. Develop simple Marks Entry Application to demonstrate accessing Database using EJB.

<Any application to the best of Teachers assumption >

8a. Develop a simple Inventory Application Using JPA.
8b. Develop a Guestbook Application Using JPA.

JPA Practical using GuestBook

Steps:
1. Create Web Application with dedicated folder for Library



2. Add Simple java class or Persitent Entity class from Database (code below GuestBook.java)
3. Add SQL Connector Jar file to Library
4. Create Persistence Unit using jdbc connection to MySQL database
5. Create the JSP files (codes given below)
6. Run the Application.

GuestBook.java
~~~~~~~~~~~~~~

```
package asif;
import javax.persistence.*;

@Entity
@Table(name="GuestBook")
public class GuestBook {
   @Id
   @GeneratedValue(strategy = GenerationType.IDENTITY)
   @Column(name="VisitorNo", unique=true, updatable=false)
   private Integer visitorNo;
   @Column(name="VisitorName")
   private String visitorName;
   @Column(name="Message")
   private String message;
   @Column(name="MessageDate")
   private String messageDate;

   public GuestBook() {
   }

   public Integer getVisitorNo() {
      return visitorNo;
   }
   public void setVisitorNo(Integer visitorNo) {
      this.visitorNo = visitorNo;
   }

   public String getVisitorName() {
      return visitorName;
```
~~~~~~~~~~~~~~

```
    }
    public void setVisitorName(String visitorName) {
        this.visitorName = visitorName;
    }

    public String getMessage() {
        return message;
    }
    public void setMessage(String message) {
        this.message = message;
    }

    public String getMessageDate() {
        return messageDate;
    }
    public void setMessageDate(String messageDate) {
        this.messageDate = messageDate;
    }
}
```

index.jsp

~~~~~~~~~~

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <body style="background-color: pink;">
         Sign the Guest Book
          <form action="GuestBookView.jsp" method="post">
                Visitor Name:  <input name="guest" maxlength="25" size="50" />
                Message: <textarea rows="5" cols="36" name="message"></textarea>
                <input type="submit" name="btnSubmit" value="Submit" />
          </form>
    </body>
</html>
```

GuestBookView.jsp

~~~~~~~~~~~~~~~~~~

```
<%@page import="java.util.*,javax.persistence.*,asif.GuestBook" %>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<%!
    private EntityManagerFactory entityManagerFactory;
    private EntityManager entityManager;
    private EntityTransaction entityTransaction;
    List<GuestBook> guestbook;
%>
<%
```

```
    entityManagerFactory = Persistence.createEntityManagerFactory("JPAApplication1PU");
    entityManager = entityManagerFactory.createEntityManager();
    String submit = request.getParameter("btnSubmit");
    if(submit != null && ("Submit").equals(submit)) {
        try {
            String guest = request.getParameter("guest");
            String message = request.getParameter("message");
            String messageDate = new java.util.Date().toString();

            GuestBook gb = new GuestBook();
            gb.setVisitorName(guest);
            gb.setMessage(message);
            gb.setMessageDate(messageDate);

            entityTransaction = entityManager.getTransaction();
            entityTransaction.begin();
            entityManager.persist(gb);
            entityTransaction.commit();
        } catch (RuntimeException e) {
            if(entityTransaction != null) entityTransaction.rollback();
            throw e;
        }
        response.sendRedirect("GuestBookView.jsp");
    }

    try {
        guestbook = entityManager.createQuery("SELECT g from GuestBook g").getResultList();
    } catch (RuntimeException e) { }
   entityManager.close();
%>
<html>
    <body>
       View the Guest Book  <b>Click <a href="index.jsp"> here</a> to sign the guestbook.</b>

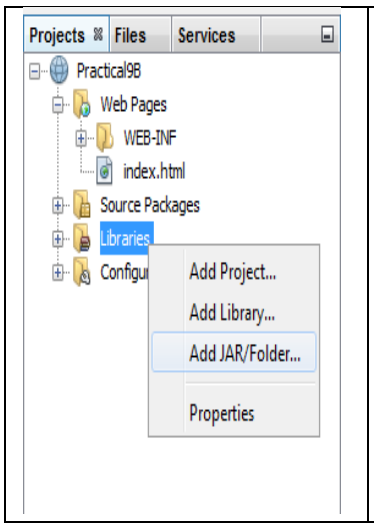
        <hr />
        <%
          Iterator iterator = guestbook.iterator();
          while (iterator.hasNext()) {
              GuestBook obj = (GuestBook) iterator.next();
         %>
             On <%= obj.getMessageDate() %>,<br />
             <b><%= obj.getVisitorName() %>:</b>
             <%= obj.getMessage() %>
             <br /><br />
         <%
          }
         %>

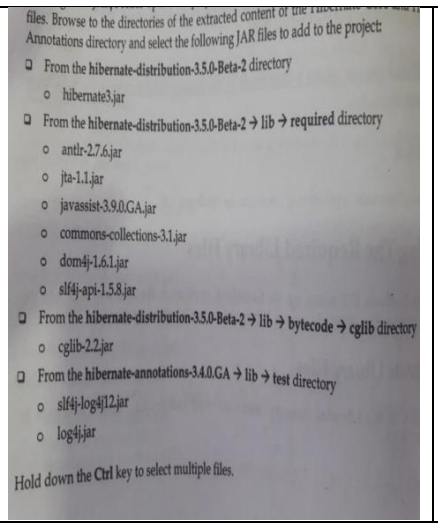
    </body>
</html>
```

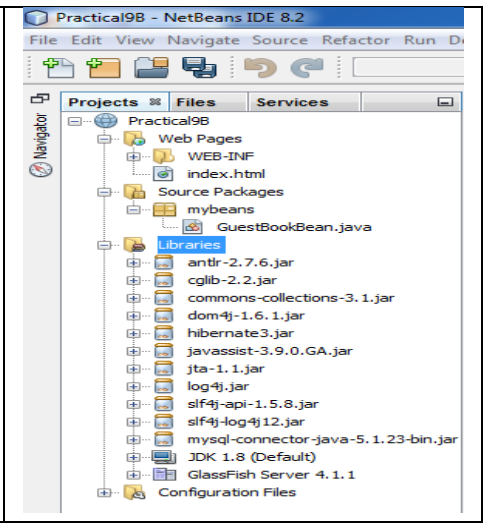
8c. Create simple JPA application to store and retrieve Book details.**<< similar to above example >>**

9a. Develop a JPA Application to demonstrate use of ORM associations.
9b. Develop a Hibernate application to store Feedback of Website Visitor in MySQL Database.

create database feedbackdb;
create table GuestBook(
vno int PRIMARY KEY AUTO_INCREMENT,
vname varchar(50),
msg varchar(100),
mdate varchar(50)
)





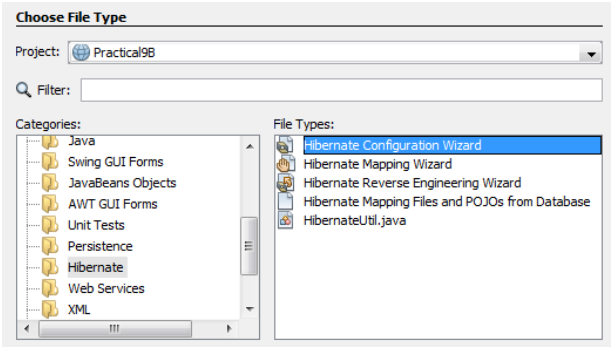


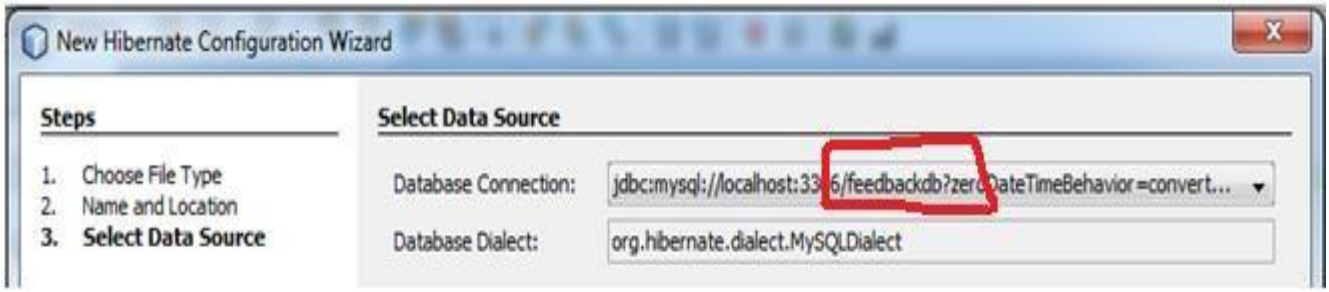
---------- GuestBookBean.java ----------

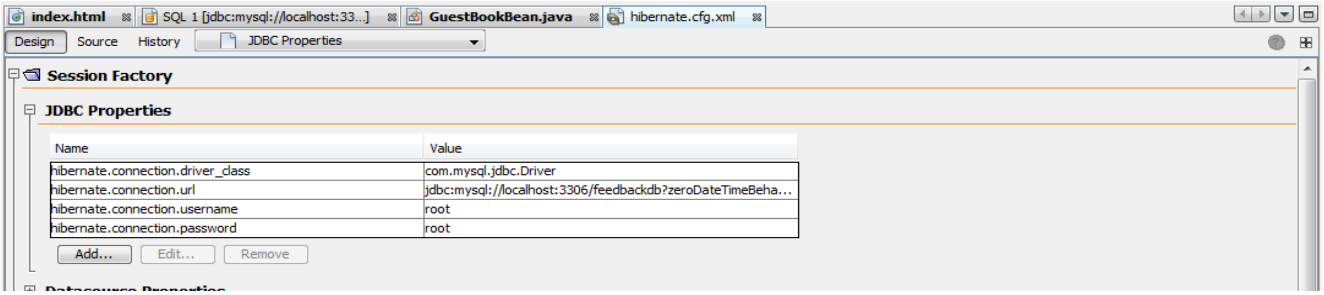
```
package mypack;
import javax.persistence.*;
@Entity
@Table(name="guestbook")
public class GuestBookBean implements java.io.Serializable {
    @Id
    @GeneratedValue
   @Column(name="vno")
   private Integer visitorNo;
   @Column(name="vname")
   private String visitorName;
   @Column(name="msg")
   private String msg;
   @Column(name="mdate")
   private String msgDate;
   public GuestBookBean() {  }
   public Integer getVisitorNo()          {    return visitorNo;  }
   public String getVisitorName()    {    return visitorName;  }
   public String getMsg()               {    return msg;  }
   public String getMsgDate()         {    return msgDate;  }
  public void setVisitorNo(Integer vn)          {    visitorNo = vn ;  }
```

```
  public void setVisitorName(String vn)     {    visitorName = vn;  }
  public void setMsg(String m)              {    msg = m;  }
  public void setMsgDate(String md)         {    msgDate=md;  }
}
```

Source packages →new → others→select category Hibernate →Hibernate Configuration Wizard








```
<hibernate-configuration>
<session-factory>
<property name="hibernate.dialect">org.hibernate.dialect.MySQLDialect</property>
<property name="hibernate.connection.driver_class">com.mysql.jdbc.Driver</property>
<property
name="hibernate.connection.url">jdbc:mysql://localhost:3306/feedbackdb?zeroDateTimeBehavior=co
nvertToNull</property>
<property name="hibernate.connection.username">root</property>
<property name="hibernate.connection.password">root</property>

<mapping class="mypack.GuestBookBean" />
</session-factory>
</hibernate-configuration>
```

---------- index.html----------

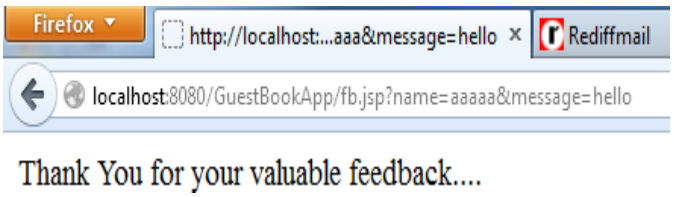
```
<h1>Website Feedback Form for google.con </h1>
<form action="fb.jsp" >
```

```
   Enter Your Name: <input type="text" name="name" ><br>
   Enter Your Message : <textarea rows="10" cols="50" name="message" ></textarea><br>
<input type="submit" value="Submit My FeedBack " >
</form>
```

---------- fb.jsp----------

```
<%@page  import="org.hibernate.*, org.hibernate.cfg.*, mypack.*"  %>
<%!             SessionFactory sf;
                org.hibernate.Session hibSession;
%>
<%
sf = new Configuration().configure().buildSessionFactory();
hibSession = sf.openSession();
Transaction tx = null;
GuestBookBean gb = new GuestBookBean();
try{
tx = hibSession.beginTransaction();
String username = request.getParameter("name");
String usermsg = request.getParameter("message");
String nowtime = ""+new java.util.Date();
gb.setVisitorName(username);
gb.setMsg(usermsg);
gb.setMsgDate(nowtime);
hibSession.save(gb);
tx.commit();
out.println("Thank You for your valuable feedback.. ");
}catch(Exception e){out.println(e);}
hibSession.close();
%>
```





9c. Develop a Hibernate application to store and retrieve employee details in MySQL Database.
<<Similar to above appication>>

10a. Develop an application to demonstrate Hibernate One- To -One Mapping Using Annotation.

10b. Develop Hibernate application to enter and retrieve course details with ORM Mapping.

10c. Develop a five page web application site using any two or three Java EE Technologies.

<<Any application to the best of teachers own assumption >>